

KHUSHI SONI

B-Tech Information Technology Undergraduate
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PROFESSIONAL SUMMARY

Software Engineering undergraduate (8.98 CGPA) with a strong analytical foundation in Python, Java, and Data Structures & Algorithms. Hands-on experience developing diverse technical solutions, including real-time web applications with API integration, IoT-based monitoring systems, and AI-driven predictive models. Proven track record of leadership as a GDG Coordinator, demonstrating a commitment to learning emerging technologies and collaborating in high-impact team environments. Eager to apply technical proficiency and research-driven problem-solving skills to build innovative, scalable software and hardware solutions.

EDUCATION

B.Tech in Information Technology

Institute of Engineering and Management, Kolkata
CGPA: 8.98/ 10 (Till 5th Semester) 2023 - 2027

Higher Secondary (Class XII)

CBSE Board - 83.4% 2021 - 2023

Secondary Education (Class X)

CBSE Board - 94.8% 2021

TECHNICAL PROFICIENCIES

- **Programming Languages:** Python, Java, C
- **Core Computer Science:** Data Structures & Algorithms, OOP, DBMS, Operating Systems, SDLC
- **Databases :** MYSQL (Relational Schema Design, SQL Queries)
- **Cloud Platform:** Google Cloud Platform (GCP), AWSLSTMs , Generative AI Prompt Engineering.
- **Tools & Platforms:** Git, GitHub, VS Code
- **Emerging Technologies :** Machine Learning, Generative AI, Cryptography , GenAI

ACADEMIC PROJECTS

Ambulance Booking System. *Flutter, Google Maps API, MySQL , Web Development*

- Developed a real-time ambulance booking platform with live location tracking and route optimization
- Integrated Google Maps API for navigation and location-based services
- Designed system workflow and user interfaces for efficient booking and request handling
- Designed database schema; backend integration currently in progress

Body Posture Detection System (IoT-Based) *Arduino, MPU6050, IoT, Web Dashboard (Planned)*

- Built an IoT-based posture monitoring system using MPU6050 sensors to detect incorrect sitting posture in real time
- Implemented sensor-based data collection and alert mechanism for posture correction
- Designed system workflow for a web-based dashboard to provide posture analytics and health insights (under development)

Energy Load Forecasting System (LSTM) *Python, TensorFlow, Pandas, Time-Series Analysis*

- Built an LSTM-based time-series model to predict hourly electricity demand using historical consumption data
 - Engineered temporal features and sequential inputs to capture daily and seasonal usage patterns
 - Achieved strong prediction accuracy, validated using MAE and loss metrics with close alignment between actual and predicted values
 - Evaluated model performance using visualization techniques and confusion matrix across demand levels
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RESEARCH PAPER (ongoing)

Hybrid Secure Predictive Routing for Payment Channel Networks (HSPR-PCN)

- Developing an AI-driven routing framework to improve transaction success rate in blockchain payment networks
 - Designing a machine learning model to predict routing success based on network parameters
 - Implementing liquidity-aware path selection to minimize transaction failures
 - Incorporating privacy-preserving mechanisms for secure and efficient transactions
 - Designing a bidding-based node selection strategy to optimize routing cost and reliability
 - Tools: Python, NetworkX, Scikit-learn
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CERTIFICATIONS

- PwC Advisory Launchpad Learning Program
 - Cyber Security Fundamentals - University of London (Coursera)
 - Classical Cryptosystems and Core Concepts - University of Colorado System (Coursera)
 - Advanced System Security Topics - University of Colorado System (Coursera)
 - Cisco Network Security: Core Security Concepts (LinkedIn Learning)
 - Cloud Computing: Understanding Core Concepts (LinkedIn Learning)
 - Introduction to Prompt Engineering for Generative AI (LinkedIn Learning)
 - Google Cloud Study Jam 2025 Completion
- Ongoing

ACHIEVEMENTS & ACTIVITIES

- Coordinator - Social Media & Outreach, GDG on Campus (IEM)
Lead the outreach strategy for technical events, including the Build On Stellar Bootcamp and Google Cloud Study Jams, resulting in increased student engagement.
- Participated in HackOasis 2.0
- Participated in ICELTS Conference